Task 1:

	CARRIER	CARRIER_NAME	UNIQUE_CARRIER	UNIQUE_CARRIER_NAME
11776	NaN	North American Airlines	NaN	North American Airlines

```
Missing CARRIER after fix: 0
```

- **CARRIER**
  - **Investigation:**
    - I examined the rows where CARRIER is missing and compared them with other carrier-identifying columns such as CARRIER_NAME, UNIQUE_CARRIER, and UNIQUE_CARRIER_NAME.
  - **Reasoning:**
    - The rows with missing CARRIER all correspond to North American Airlines. The carrier code "NA" was interpreted by pandas as a missing value (NaN) when reading the file. Since the airline identity is clearly known from other columns, this is a data-loading issue rather than genuine missingness.
  - **Imputation decision:**
    - Restored CARRIER = "NA" for rows where CARRIER_NAME is "North American Airlines". This is better described as a correction of misread data than a statistical imputation.

	CARRIER	CARRIER_NAME	AIRLINE_ID	UNIQUE_CARRIER	UNIQUE_CARRIER_NAME
11465	L4	NaN	NaN	NaN	NaN
54610	OH	NaN	NaN	NaN	NaN

```
Missing CARRIER_NAME after imputation: 0
Missing AIRLINE_ID after imputation: 0
```

- **CARRIER_NAME**
  - **Investigation:**
    - Examined rows where CARRIER_NAME is missing and compared them to other rows with the same
  - **Reasoning:**
    - For a given carrier code, the carrier name should be consistent across the dataset. Since non-missing rows with the same CARRIER exist, the missing value can be recovered from them.
  - **Imputation decision:**

        - Imputed missing CARRIER_NAME using the most common observed CARRIER_NAME among rows with the same CARRIER.
- **AIRLINE_ID**
    - **Investigation:**
        - I inspected the rows with missing AIRLINE_ID and compared them to other rows sharing the same CARRIER code.
    - **Reasoning:**
        - AIRLINE_ID is a carrier identity field, so it should usually be consistent for a given carrier in the dataset. Since the same carrier appears in other rows with non-missing AIRLINE_ID, the missing values can be recovered from those matching rows.
    - **Imputation decision:**
        - I imputed missing AIRLINE_ID values using the most common observed AIRLINE_ID among rows with the same CARRIER.

	MANUFACTURER	MODEL	MANUFACTURE_YEAR
116651	BOEING	B747-400	NaN
124680	BOMBARDIER	CL-600-2C10	NaN
124681	BOMBARDIER	CL-600-2C10	NaN

```
Missing MANUFACTURE_YEAR after imputation: 0
```

- **MANUFACTURE_YEAR**
    - **Investigation:**
        - I checked the rows where MANUFACTURE_YEAR is missing and looked at the associated aircraft MODEL.
    - **Reasoning:**
        - Aircraft of the same model tend to come from a similar production period, so the model provides useful information for estimating a missing manufacture year. Because only a small number of rows are missing this value, a simple model-based imputation is reasonable. Using the median within MODEL is more robust than using the mean because it is less affected by unusually old or new aircraft.
    - **Imputation decision:**
        - I imputed missing MANUFACTURE_YEAR values using the median manufacture year within each MODEL, and used the global median only as a fallback if a model-level value was unavailable.

```
Missing rate for NUMBER_OF_SEATS: 0.1000506375034955
Missing rate for CAPACITY_IN_POUNDS: 0.100761074119701
```

Seat missingness by AIRCRAFT_TYPE:

AIRCRAFT_TYPE	SEATS_MISSING
6871	1.0
6961	1.0
699	1.0
6371	1.0
6231	1.0
...	...
7461	0.0
7481	0.0
2162	0.0
1502.0	0.0
6791	0.0

150 rows × 1 columns

Capacity missingness by AIRCRAFT_TYPE:

AIRCRAFT_TYPE	CAPACITY_MISSING
7886	1.0
699	1.0
723	1.0
724	1.0
677	1.0
...	...
6751	0.0
6761	0.0
6711.0	0.0
6861	0.0
6871	0.0

150 rows × 1 columns

Seat missingness by MODEL:

MODEL	mean	count
767-332ER-PSGR	1.0	546
B777-222BPAX	1.0	440
B777-2	1.0	380
B767-323ERPASSENGER	1.0	290
B767-322ERPAX	1.0	258
B757-300PAX	1.0	252
B777-200PASSENGER	1.0	236
A330-323-PSGR	1.0	235
767-432ER-PSGR	1.0	231
B777-222PAX	1.0	228
B777-200PAX	1.0	220
A330-243	1.0	210
A330-200	1.0	201
A-321-PSGR	1.0	198
B747-422PAX	1.0	194

Capacity missingness by MODEL:

MODEL	mean	count
737-7H4	1.0	569
737-8	1.0	531
ERJ170-200LR	1.0	348
737-823	1.0	277
737-824	1.0	270
737-924ER	1.0	263
737-8H4	1.0	256
CL-600-2D24	1.0	247
208B	1.0	225
CL-600-2C10	1.0	219
737-9	1.0	127
A319-131	1.0	109
777-222	1.0	104
787-9	1.0	98
A320-271N	1.0	92

```
Missing NUMBER_OF_SEATS after imputation: 0
Missing CAPACITY_IN_POUNDS after imputation: 0
```

- **NUMBER_OF_SEATS** (missingness category = MAR)
  - **Investigation:**

        - I created an indicator for whether NUMBER_OF_SEATS is missing and compared the missingness rate across observed variables such as AIRCRAFT_TYPE and MODEL.
    - **Ruling out MCAR:** Under MCAR, missing values would be uniformly distributed — we would expect roughly the same ~10% rate in every AIRCRAFT_TYPE and MODEL group. Instead, certain aircraft types show 100% missingness while others show 0%. This non-uniform pattern rules out MCAR.
    - **Ruling out MNAR:** MNAR would mean the probability of missingness depends on the actual unobserved seat count itself (example: aircraft with very high or very low seat counts are more likely to have missing values). There is no evidence of this, as missingness clusters by aircraft type and model, not by any particular range of seat values. The pattern is better explained by reporting practices tied to aircraft type (an observed variable), not the seat count causing its own missingness.
    - **Conclusion (MAR)**: Missingness depends on observed variables (AIRCRAFT_TYPE, MODEL), which is the definition of MAR. This justifies using model-based imputation.
    - **Imputation decision:**
        - I imputed missing NUMBER_OF_SEATS values using the median number of seats within each MODEL, with the overall median as a fallback.
- **CAPACITY_IN_POUNDS** (missingness category = MAR)
    - **Investigation:**
        - I created an indicator for whether CAPACITY_IN_POUNDS is missing and compared the missingness rate across observed variables such as AIRCRAFT_TYPE and MODEL.
    - **Ruling out MCAR:** The same non-uniform pattern observed for NUMBER_OF_SEATS applies here: certain aircraft types have 100% missing capacity values while others have none. A uniform ~10% rate across all groups would be expected under MCAR, which is not observed.
    - **Ruling out MNAR:** MNAR would imply that aircraft with unusually high or low capacity are more likely to have missing capacity values. There is no evidence of this, as missingness clusters by aircraft type and model, not by any particular range of capacity values. A reporting or data entry gap specific to certain aircraft types (an observed variable) is a more plausible explanation than the capacity value itself causing its own missingness.
    - **Conclusion** (MAR): Missingness depends on observed aircraft characteristics (AIRCRAFT_TYPE, MODEL), consistent with MAR.
    - **Imputation decision:**
        - I imputed missing CAPACITY_IN_POUNDS values using the median capacity within each MODEL, with the global median as a fallback.

Task 2:

MANUFACTURER	count
BOEING	15922
Embraer	11508
THEBOEINGCO	9223
Bombardier	8871
Boeing	8392
BoeingCo	7446
AIRBUS	7179
AirbusIndustries	6967
BOEINGCOMPANY	6767
Airbus	5289

MANUFACTURER	count
BOEING	48750
AIRBUS	23513
BOMBARDIER	16344
EMBRAER	15554
MCDONNELL-DOUGLAS	9506
BOEINGCOMPANY	6767
CESSNA	4514
ATR	1181
DOUGLAS	1137
GE	1110
DE HAVILLAND	1084
GULFSTREAM	441
SAABAIRCRAFT	411
CONVAIR	208

- MANUFACTURER
  - **Is transformation or standardization required?**
    - Yes. Standardization is required.
  - **Reasoning:**
    - I inspected the MANUFACTURER column using value_counts() and observed that the same manufacturer appears under multiple variations due to differences in capitalization, punctuation, spacing, and naming conventions (i.e., “BOEING”, “Boeing Co”, “THE BOEING COMPANY”). These variations refer to the same entity but would be treated as distinct values without cleaning, leading to incorrect aggregation and analysis.
  - **Transformation decision:**
    - I standardized the MANUFACTURER column by converting all values to uppercase, removing punctuation and excess whitespace, and mapping common variants to canonical names (i.e., mapping Boeing variants to “BOEING”, Airbus variants to “AIRBUS”). This reduces duplication and ensures consistency across the dataset.

MODEL	count
EMB-145	2614
B-737-7H4	2470
B737-823	2370
A320-232	2333
A321-231	2259
737-700PASSENGERONLY	2027
C-208B	1872
B757-2	1775
CRJ-2/4	1761
B737-800PAX	1621
MD-80	1610
ERJ-170-200LR	1379
757-200	1345
CRJ200-2B19	1342
A319	1267
B-737-8H4	1256
CRJ-200	1148

MODEL	count
B737-823	3326
EMB145	3052
CRJ-200	2790
B737-7H4	2474
A320-232	2463
B757-2	2369
A321-231	2299
737-700	2044
CRJ-900	2011
B737-800	1909
ERJ-170-200LR	1886
C-208B	1872
CRJ-2/4	1761
A319	1725
ERJ-175	1692
A320	1613
MD-80	1611

- MODEL
  - **Is transformation or standardization required?**
    - Yes. Moderate standardization is required.
  - **Reasoning:**
    - I inspected the MODEL column using value_counts() and observed substantial inconsistency. The same aircraft models appeared under multiple formats due to differences in capitalization, spacing, punctuation, and prefix usage (i.e., “B-737” vs “B737”, “CRJ200” vs “CRJ-200”). In addition, many entries contained non-informative suffixes such as “PASSENGER”, “PAX”, and “PSGR”, which seem to describe usage rather than the model itself and create unnecessary duplication.
    - I also observed cases where the manufacturer name was concatenated with the model (i.e., “AIRBUSA320-232”), which led to artificial distinctions between otherwise identical models.
    - At the same time, many differences in model names represent meaningful distinctions (i.e., “B737-700” vs “B737-800”), so aggressive merging might risk combining different aircraft types.
  - **Transformation decision:**
    - I applied moderate standardization by:
      - converting all values to uppercase and trimming whitespace
      - removing common non-informative suffixes (i.e., “PASSENGER”, “PAX”, “PSGR”)
      - normalizing common formatting patterns (i.e., standardizing “B-737” to “B737” and “CRJ200” to “CRJ-200”)

- removing cases where manufacturer names (i.e., “AIRBUS”) were concatenated with the model.
- I avoided merging core model identifiers to preserve meaningful distinctions between aircraft types.

AIRCRAFT_STATUS	count
O	79487
b	30852
B	12699
a	7804
A	1330
L	122
o	19

AIRCRAFT_STATUS	count
O	79506
B	43551
A	9134
L	122

- AIRCRAFT_STATUS
  - **Is transformation or standardization required?**
    - Yes. Standardization is required.
  - **Reasoning:**
    - I inspected the AIRCRAFT_STATUS column using value_counts() and found that it consists of single-letter categorical codes (i.e., “O”, “A”, “B”, “L”). However, the same categories appear in both uppercase and lowercase forms (i.e., “B” and “b”, “A” and “a”, “O” and “o”), which would cause them to be treated as different categories.
  - **Transformation decision:**
    - I standardized the column by converting all values to uppercase and removing any extra whitespace. This ensures that each category is represented consistently.

OPERATING_STATUS	count
Y	126577
N	5664
y	71
	1

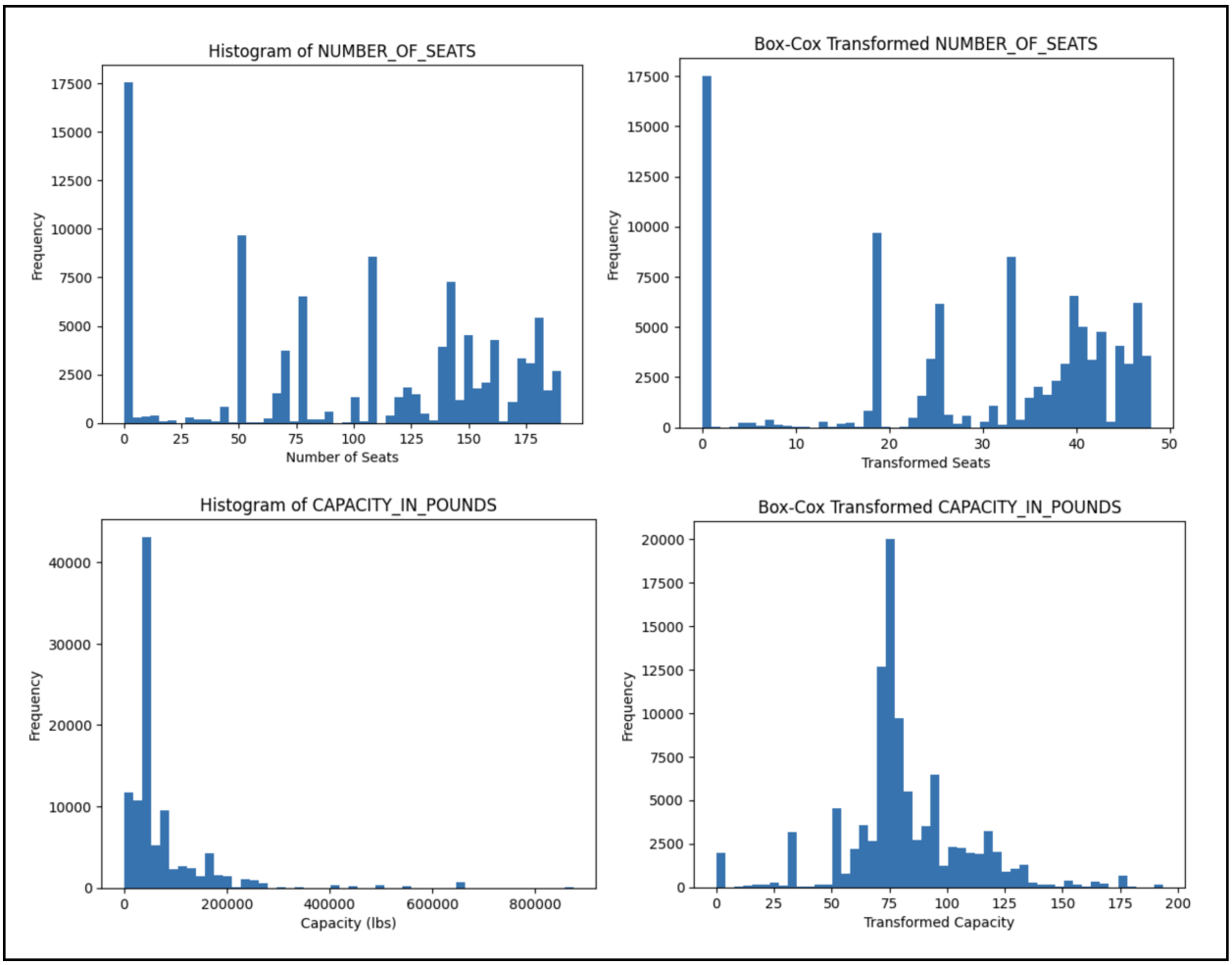
OPERATING_STATUS	count
Y	126649
N	5664

- OPERATING_STATUS
  - **Is transformation or standardization required?**
    - Yes. Standardization is required.
  - **Reasoning:**
    - I inspected the OPERATING_STATUS column using value_counts() and found that it contains categorical values ("Y" and "N"), along with a small number of lowercase entries ("y"), indicating inconsistent formatting. In addition, there is a small number of missing values in the column.
  - **Transformation decision:**
    - I standardized the column by converting all values to uppercase and removing any extra whitespace to ensure consistency. I also filled the missing value using the mode (most frequent value), since the column is binary and overwhelmingly dominated by one category. This results in a consistent and complete representation of the variable.

Task 3:

After removing all rows with remaining missing values, the dataset contains 101287 rows. In total, 31026 rows were removed during this step.

Task 4:



The skewness of **NUMBER_OF_SEATS** is -0.27, indicating a roughly symmetric distribution with a slight left skew. This suggests that the data are fairly balanced, with a mild tendency toward higher seat counts rather than a long right tail.

The skewness of **CAPACITY_IN_POUNDS** is 4.31, indicating a highly right-skewed distribution. This suggests that while most aircraft have relatively low to moderate capacity, a small number of aircraft have extremely large capacities, creating a long right tail.

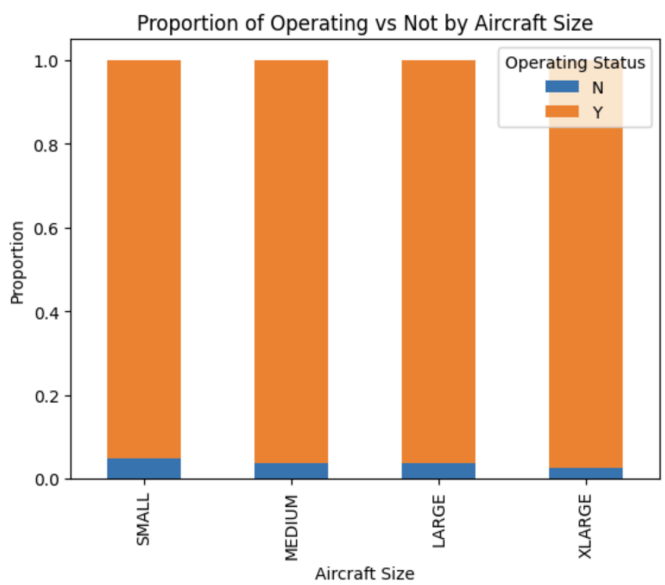
Before transformation, the distribution of **NUMBER_OF_SEATS** shows a large spike at 0 and a spread of values across typical seat counts, with no strong overall skew but noticeable clustering. After applying the Box-Cox transformation, the distribution remains largely similar in shape, with only minor compression of the scale. The transformation does not significantly improve normality, primarily because the data is not heavily skewed and the large concentration of zero values continues to dominate the distribution.

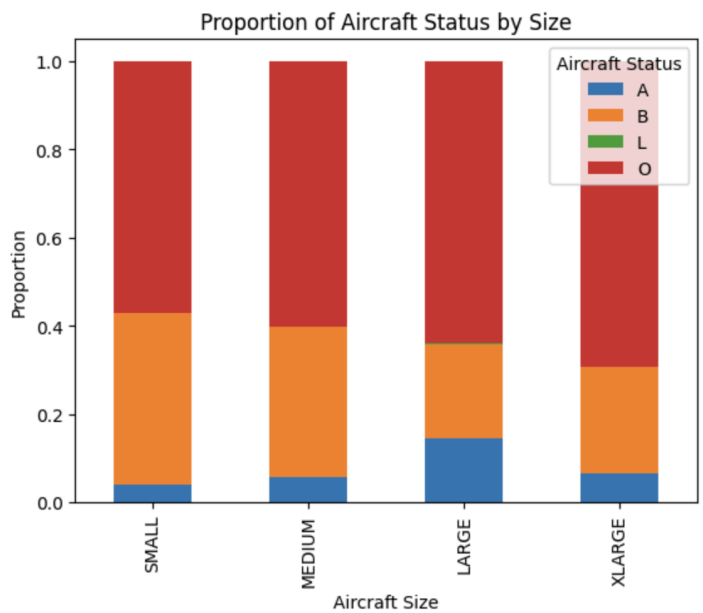
Before transformation, the distribution of **CAPACITY_IN_POUNDS** is highly right-skewed, with most observations concentrated at lower values and a long tail extending toward very large capacities. After applying the Box-Cox transformation, the distribution becomes more symmetric and takes on a more

bell-shaped form, with the right tail significantly compressed. This indicates that the transformation makes the data more normal-like and reduces the influence of extreme values.

Task 5:

OPERATING_STATUS	N	Y
SIZE		
SMALL	0.048170	0.951830
MEDIUM	0.036717	0.963283
LARGE	0.037526	0.962474
XLARGE	0.027129	0.972871





Across all aircraft size groups, the vast majority of aircraft are operating, with over 94% operating in every category. There is a slight upward trend, where larger aircraft are more likely to be operating, increasing from about 94% for small aircraft to nearly 97% for extra-large aircraft.

Looking at aircraft status, “O” is the most common category across all size groups and becomes more dominant as aircraft size increases. The proportion of “B” decreases steadily as size increases, indicating that smaller aircraft are more likely to fall into this category. The proportion of “A” increases from small to large aircraft but slightly decreases for extra-large aircraft, while “L” remains negligible across all groups.

Overall, larger aircraft tend to be more consistently operational and are more concentrated in the primary “O” status category, while smaller aircraft exhibit slightly more variation in status.

Task 6:

	train_RMSE	test_RMSE
LinearRegression_NUMBER_OF_SEATS	59.45	59.11
RandomForest_NUMBER_OF_SEATS	2.06	2.22
LinearRegression_CAPACITY_IN_POUNDS	88866.54	89674.48
RandomForest_CAPACITY_IN_POUNDS	29263.20	32428.60

Random forest substantially outperforms linear regression for both targets. For **NUMBER_OF_SEATS**, linear regression has a test RMSE of 59.11, while random forest brings it down to 2.22, a dramatic improvement. For **CAPACITY_IN_POUNDS**, linear regression has a test RMSE of 89674 lbs vs 32429 lbs for random forest.

Both linear regression models show nearly identical train and test RMSE (~59 and ~88867 respectively), which indicates the model is too simple to capture the structure in the data (high bias, low variance).

The random forest shows a small gap between train and test RMSE for both targets (2.06 vs 2.22 for seats; 29263 vs 32429 for capacity), suggesting mild overfitting but strong generalization overall.

CAPACITY_IN_POUNDS remains harder to predict than **NUMBER_OF_SEATS** for both models, likely because aircraft capacity varies more continuously and is influenced by configuration factors not fully captured by the available features.

AI Usage Note: (1) tool(s) used, (2) the key prompt(s), and (3) what you changed and how you verified the results.

1. Task 2

- ChatGPT
- Prompt:
    - this is the output when i initially inspect MODEL EMB-145 2614 B-737-7H4 2470 B737-823 2370 A320-232 2333 A321-231 2259 737-700PASSENGERONLY 2027 C-208B 1872 B757-2 1775 CRJ-2/4 1761 B737-800PAX 1621 MD-80 1610 ERJ-170-200LR 1379 757-200 1345 CRJ200-2B19 1342 A319 1267 B-737-8H4 1256 CRJ-200 1148 ERJ-175 1132 SUPER80PASSENGER 1108 A320-1/2 1107 757-24APF 1039 B737-3 1036 MD-88-PSGR 1028 C-208B/3 1017 MD-11 991 757-232-PSGR 988 CRJ-900LR-PSGR 976 B737-823PASSENGER 956 B737-8 951 B737-7/L 929
    - Can you help me understand how these plane models are supposed to be grouped together?
- Through a google search a verified that its response about things like the suffixes of the plane model titles like 800PAX, 823PASSENGER, etc were true. As specified in the instructions, it is not needed to catch all the inconsistencies, so I chose to use this information to moderately standardize.